

trolled by a microcontroller; in case of TRN-1, an Arduino Mega 2560 board. You can also drive it by connecting it to your PC or laptop.

"Using an Arduino board provides a good number of inputs/outputs (useful for attaching a number of sensors) and is also easy to program, highly stable and extremely flexible. Stepper motors can easily be replaced with DC motors or servo motors, and all of them can be controlled directly by the board without much fuss," Varkey says.

The Arduino board is capable of driving up to eight axes (eight stepper motors) at the same time. Hence your system has around eight degrees of freedom. You can add a proximity sensor, IR sensor or whatever suits you. The different motors are connected to each other to help the robot move the way you like.

Each of the three parts of the kit can be customised to a great degree, enabling smooth functioning and lending great frequency to the robot that you want to design.

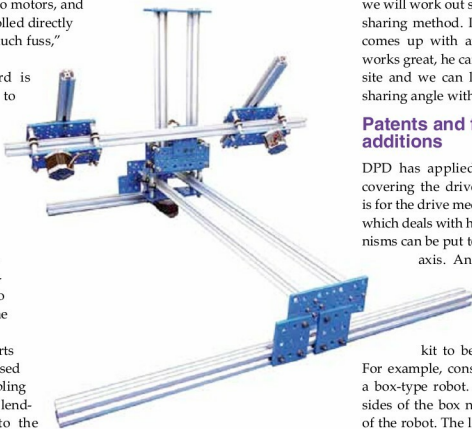
Obstacles

Talking about the problems encountered during the materialisation of the idea, Varkey says, "Since the idea was already in my head after the visits to engineering colleges, I spent a lot of time listening to the target users of our system, making notes of the problems they were facing, what they wanted and things they wanted in the solution. Then, it took us nearly a year to reach the point where we are today. This model is actually based on the 10th prototype we rolled out. The first one we made could only be turned into a single basic robotic system."

"To add value, Merlyn TRN-1 was worked on extensively to enable it to be turned into 28 fundamentally dif-

ferent types of robotic systems. You can cascade different kits together to increase the envelope in which you are working along with the range/scope of the work you are doing. The best part is that you are not limited by the mechanical, electronics or software scheme of things utilised in the robot," he adds.

The components were especially



made from anodised aluminium, ensuring a rugged yet light design.

Spreading the wings

The kit was premiered at the EFY Expo 2013, held recently in New Delhi, as a project concept for students.

Varkey shares, "A trial run for the product has already been done at some engineering colleges by way of live demonstrations, and almost all of those colleges have placed orders for our product. They said that this is something very new and they would love to have such a useful product in their college."

Varkey plans to market the product by directly contacting the engineering institutes, through dealers who are already selling kits in the market, and

by seeking help of media outlets like Electronics For You and the likes to spread word about this product.

Varkey adds, "Another way we have thought of is by utilising the open source way. Our kit is open for all sorts of customisation. If a software guy comes up with a neat applet that can be integrated into the robot to perform some particularly specific function, we will work out some sort of revenue sharing method. If a mechanical guy comes up with an end effector that works great, he can put it on our website and we can look at the revenue sharing angle with him as well."

Patents and future additions

DPD has applied for three patents covering the drive mechanisms. One is for the drive mechanism being used, which deals with how different mechanisms can be put together on the same

axis. Another patent is for the direct drive mechanism that allows any linear member of the

kit to become a linear axis.

For example, consider the making of a box-type robot. In this case, the 12 sides of the box now become 12 axes of the robot. The last one is regarding the specific drive mechanism where you can use any member of the kit and drive it directly through the stepper motor without using any type of belt. In the system, you can have a linear drive with the stepper motor using a screw-drive while having a DC motor on another axis, which has a wire-drive to move around—all of them independently working together on different fields and different actuators.

Regarding additions to the design, Varkey says, "Adding new features is the basis of all innovations and we at DPD are no different. We have plans to increase the amount of customisation available and improve the graphical user interface of the microcontroller and a host of other features." ●

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